

Indian Institute of Information Technology Vadodara - International Campus Diu



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Branch: **CSE**
Batch: **2023**
Subject: **CS 268**

Aim:

Design and implementation of a register bank in Verilog.

Theory:

Registers are the basic building block of any hardware design. The size of instructions executed by the processor depends on the size of registers.

Objective:

Design & simulate a register bank which has 2 register reads and 1 register write in every clock cycle. There is an assumption that the same register can not be read and written at the same time.

Verilog Code:

regbank.v

```
module regbank(rdata1, rdata2, wrdata, rs1, rs2, dr, write, reset, clock);  
    input [4:0] rs1, rs2, dr;  
    input clock, write, reset;  
    output [31:0] rdata1, rdata2;  
    input [31:0] wrdata;  
    reg [31:0] regbank1[0:31]; // Register bank of 32 registers, each 32-bit wide  
    integer k;  
  
    assign rdata1 = regbank1[rs1];
```

```

    assign rdata2 = regbank1[rs2];

    always @(posedge clock) begin
        if (reset) begin
            for (k = 0; k < 32; k = k + 1)
                regbank1[k] <= 0;
        end else if (write) begin
            regbank1[dr] <= wrdata; // Fixed write operation
        end
    end
end
endmodule

```

regbank_tb.v

```

module regbank_tb();

    reg [4:0] rs1, rs2, dr;
    reg clock, write, reset;
    wire [31:0] rdata1, rdata2;
    reg [31:0] wrdata;
    integer k;

    regbank REG(rdata1, rdata2, wrdata, rs1, rs2, dr, write, reset, clock);

    initial clock = 0;
    always #5 clock = ~clock;

    initial begin
        $dumpfile("testregbank32.vcd");
        $dumpvars(0, regbank_tb);

        // Reset the register bank
        #1 reset = 1; write = 0;
        #5 reset = 0;

        // Writing values to registers
        #7 for (k = 0; k < 32; k = k + 1) begin
            dr = k;
            wrdata = 10 * k;
        end
    end
endmodule

```

```

        write = 1;

        #10 write = 0;

    end

    // Reading values from registers

    #20 for (k = 0; k < 32; k = k + 2) begin

        #10 rs1 = k;

        rs2 = k + 1;

        $display("reg[%2d] = %d, reg[%2d] = %d", rs1, rdata1, rs2, rdata2);

    end

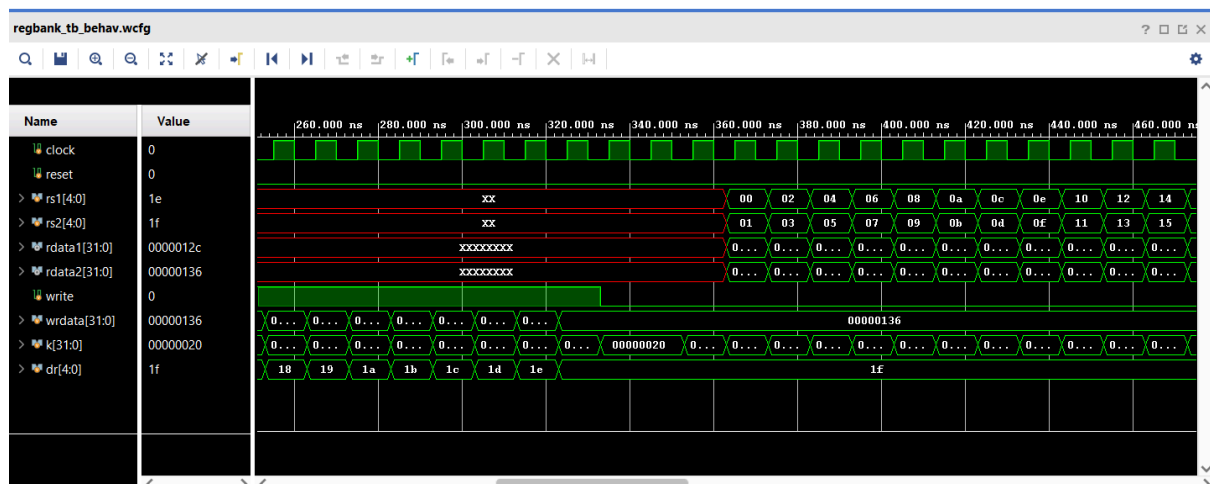
    #500 $finish;

end

endmodule

```

Vivado Output:



Schematic Diagram:

1090 Cells

114 I/O Ports

1203 Nets

